Capitalizing on digital financial services for clean cooking transitions: Evidence from GasPay in Ghana

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Abstract

Over 2 billion people worldwide still rely primarily on polluting cooking fuels. These households often have intermittent incomes and purchase firewood or charcoal day-to-day. In many low and middle income countries, liquefied petroleum gas (LPG) is the main feasible clean fuel, but requires larger payments to refill or exchange cylinders. Even when LPG's total cost matches biomass fuels, liquidity constraints can prevent households from switching. GasPay—a phone-based digital financial service (DFS) platform connecting users with LPG suppliers while providing a designated savings vehicle for refills—addresses this barrier. We describe the platform's design, the motivation behind it, and estimate its potential impact on charcoal demand, LPG use, and carbon emissions. Our results reveal how DFS might unlock clean energy access and advance multiple development goals simultaneously.

Introduction

Over two billion individuals globally cook with smoky, solid fuels, resulting in devastating health and climate impacts (1). The unaffordability of clean cooking fuels such as liquefied petroleum gas (LPG) remains one of the largest barriers to their adoption and sustained use in low and middle income countries (LMICs) (1). Financial tools could alleviate clean fuel unaffordability, yet many individuals in LMICs—particularly women—lack access to such services.

Digital financial services (DFS), specifically mobile money platforms, have emerged to provide financial tools, allowing users to store and transfer monetary value with low-cost mobile phones. Associated platforms with smart phone applications or unstructured supplementary service data (USSD) “short codes” allow users to send money to friends, family members, and businesses. However, DFS has never been applied to cooking fuel purchases. A partnership including private financial service providers, software developers, government and academic researchers developed GasPay, a phone-based DFS platform in peri-urban Ghana that enables users to securely and conveniently save toward LPG refills, earn savings incentives, purchase LPG, access microloans, and schedule deliveries.

This Report describes the financial challenge, our theory of change based on a national survey, focus groups and interviews, and the platform we designed, GasPay. We estimate the potential near term scale of GasPay and its implications for fuel demand and climate. Finally, we discuss how our results might advance other development objectives.

Results

The problem

Annually, polluting cooking fuel use contributes to 2.8 million premature deaths from household air pollution (2) and 1.1 gigatonnes of carbon dioxide equivalent emissions (3). Further, main cooks, typically women, must spend hours a week collecting or purchasing fuel and then cooking for longer time periods than required by more efficient modern fuels (4). Clean fuels, the most readily available of which in many LMICs is LPG, reduce the risk of disease (5), carbon emissions (3), and opportunity costs of time (6). However, the unaffordability of clean fuels prevents their adoption and the realization of these health, climate, and well-being benefits.

Clean fuels pose a financial challenge to users in terms of aggregate expenditure as well as liquidity (cash on hand). Users must afford both the upfront cost of the stove and the total fuel cost, often requiring a lump sum given available fuel quantities.

Evaluating expenditure on cooking fuel through a 2021 national household energy use survey of Ghana, we find that exclusive charcoal and LPG users spent 2.3 (sd:0.14) and 3.6 (sd: 0.35) 2025 USD per standard adult equivalent, respectively, on cooking fuel in the last month. Individuals who used primarily charcoal and secondarily LPG (and vice versa) spent 3.4 (sd: 0.43) USD and 3.2 (sd: 0.31) per standard adult equivalent last month on cooking fuel. While expenditure alone does not reflect consumption and household sizes vary, polluting and clean fuel users may face more comparable aggregate cooking energy costs than previously thought (7, 8).

Despite similar expenditures, liquidity constraints remain. Polluting fuel users often have intermittent incomes and purchase small quantities of firewood or charcoal (7). LPG requires a larger lump sum for the stove and fuel. Firewood and charcoal can be portioned off into various quantities, while LPG must be purchased in specific cylinder sizes. The smallest purchasible quantity of charcoal costs about 0.25 USD, while the smallest widely available LPG cylinder refill costs 8 USD. Under a cylinder recirculation model, households must use the entire cylinder and exchange the empty cylinder for a full one. Thus, even if households are spending the same amount on fuel in aggregate, clean fuels may remain unaffordable if users do not have the full cylinder cost all at one time. Without access to microloans or accounts to save, liquidity constraints can prevent clean fuel use (7).

Our theory of change

We hypothesize DFS can reduce liquidity constraints along with transaction and opportunity costs, thereby promoting clean fuel use. In theory, DFS could provide individuals with a safe, convenient account to redirect incremental amounts previously spent on charcoal towards savings for their LPG refill. The DFS could incentivize LPG purchases or savings behaviour through bonuses, subsidies, or loyalty points. Additionally, users who are short on the cash needed for the LPG exchange could receive a microloan to avoid reverting back to polluting fuel use. We base these hypotheses in the literature on liquidity constraints for clean fuels (7) and results from our own focus groups and interviews discussing these potential features. We envision that DFS will be the most relevant to charcoal users, as firewood, another prominent dirty fuel, can be purchased but is often 'freely' collected at the cost of women's time.

A clean cooking fuel specific DFS could also reduce transaction costs of LPG purchases, as well as the time burden associated with purchasing cooking fuel frequently. LPG refill stations are often far from a users' home and transporting cylinders is difficult. A DFS would allow users to check stock and schedule a delivery. Beyond the transaction, LPG would save users time, allowing for additional household chores, leisure, or economic activities.

Finally, we envision that this DFS could also support focused targeting of subsidies or incentives. Eligibility for these benefits could relate to need and/or on users during critical windows of vulnerability to air pollution (e.g., pregnancy).

The GasPay platform

After extensive household surveys and focus groups, we developed GasPay, a mobile money based DFS, where consumers can register for a wallet, make savings deposits, check their balances, order LPG, and give feedback. The platform allows consumers to earn bonuses for saving and obtain small advances for LPG. The GasPay platform works through USSD messaging and an application for smartphones.

LPG companies can set up LPG retail locations throughout communities and assign employees, add inventory, create promotions, and track sales. Suppliers can also review and download business analytics and receive consumer feedback. The platform is built to host multiple LPG providers.

We are currently conducting a randomized control trial with roughly 1,800 users across Techiman, Bono East, Ghana to understand the causal impacts of this platform on LPG and polluting fuel use. Simultaneously, we are working to commercialize GasPay to increase the customer base and reach within Ghana.

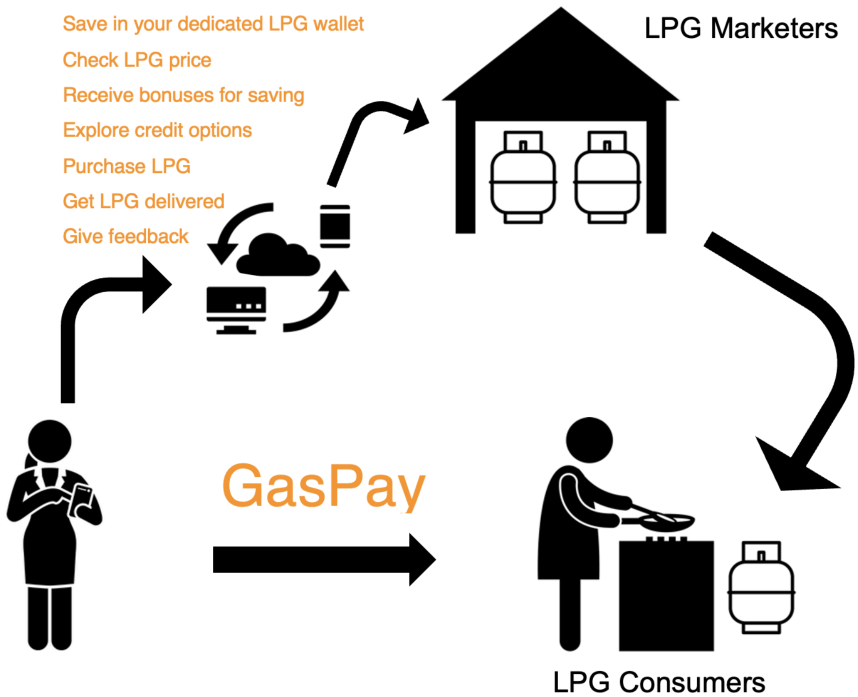

Figure 1. GasPay model

The potential scale

For Ghana, we outline the user types likely to be most responsive to GasPay (Figure 1). In principle, all current expenditure on purchased dirty fuels could be redirected to LPG, provided users can overcome liquidity barriers through small savings. Thus, we project the potential scale of GasPay to be the number of primary and secondary charcoal users in Ghana who may now be able to transition to LPG via relaxed liquidity constraints through GasPay.

Despite challenges with survey weights, limiting full national representativeness, the 2021 survey found that 24% of individuals use charcoal exclusively, while 14% primarily use LPG with some charcoal and 6.9% primarily use charcoal with some LPG. If these primary and secondary charcoal users in Ghana transition to LPG via GasPay, annual LPG demand could increase nationally by 0.19 megatonnes (Mt) [95% Confidence Interval (CI): 0.15, 0.22] of LPG, replacing 0.56 Mt [95% CI: 0.47, 0.66] of charcoal. If GasPay could transition all residential charcoal use to LPG by relaxing liquidity and transaction costs, the rate of any LPG use could increase from ~36% of Ghanaian households (as of 2021) to ~75% (Figure 2).

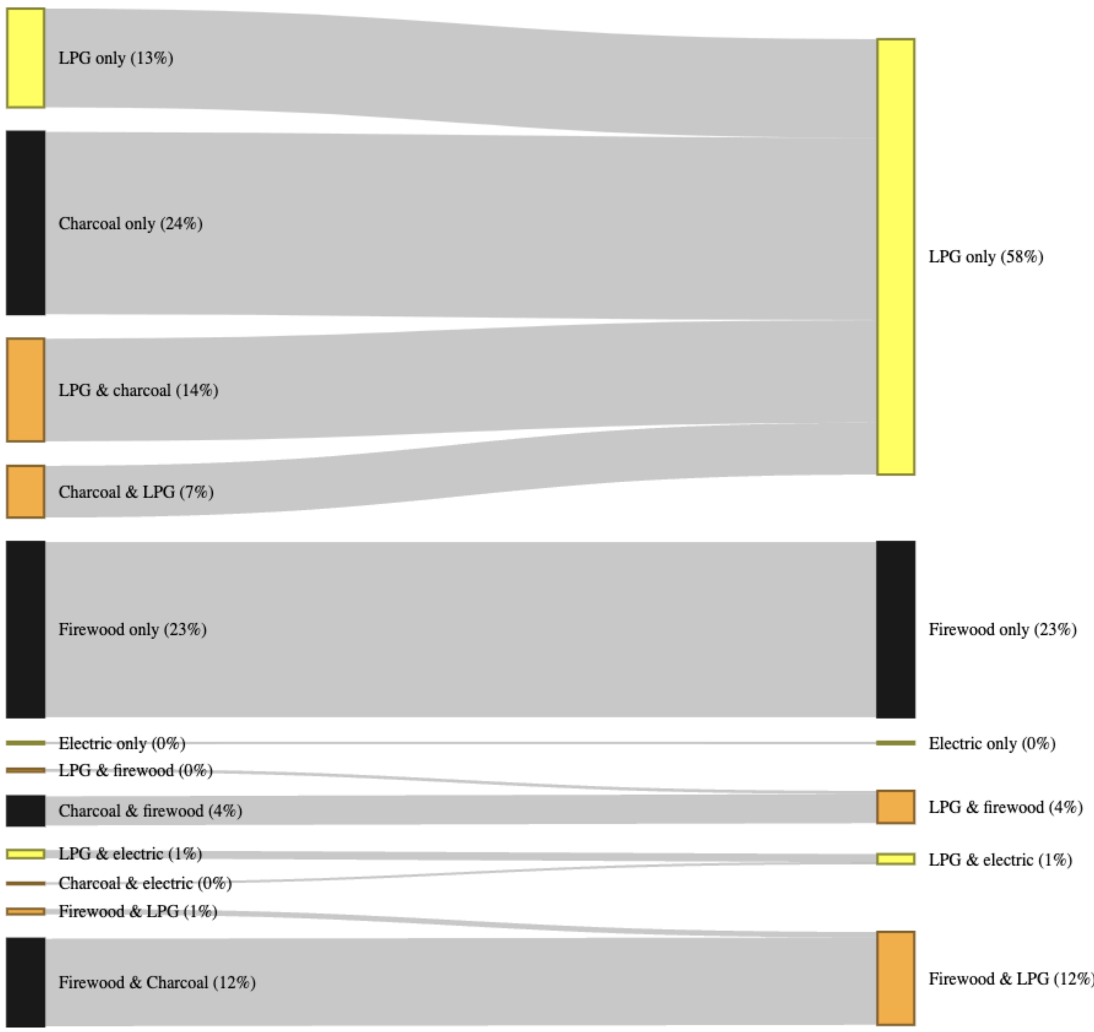

Figure 2. This sankey diagram indicates the potential transition GasPay, a digital financial service platform, could facilitate (left to right). Using a national survey of Ghana, we assume that all expenditure on purchased charcoal could be redirected to LPG with the GasPay platform through eased liquidity constraints. This excludes the minor transitions of electricity users.

Given the 2021 population of ~31 million and charcoal demand, this additional LPG use and avoided charcoal use could avoid 2.4 Mt [95% CI: 1.9, 2.7] of carbon dioxide equivalent per year, accounting for carbon dioxide, methane, carbon monoxide, and nitrous oxide and the fraction of non-renewable biomass (see Methods). This represents 10% of Ghana's total estimated emission for 2021 (9). Further, these emission reductions are likely underestimated for 2025 given growth in population and charcoal demand. Peri-urban and urban settings show the highest potential for impact, where the largest number of charcoal users reside, and where LPG supply infrastructure is already developed (Figure 3).

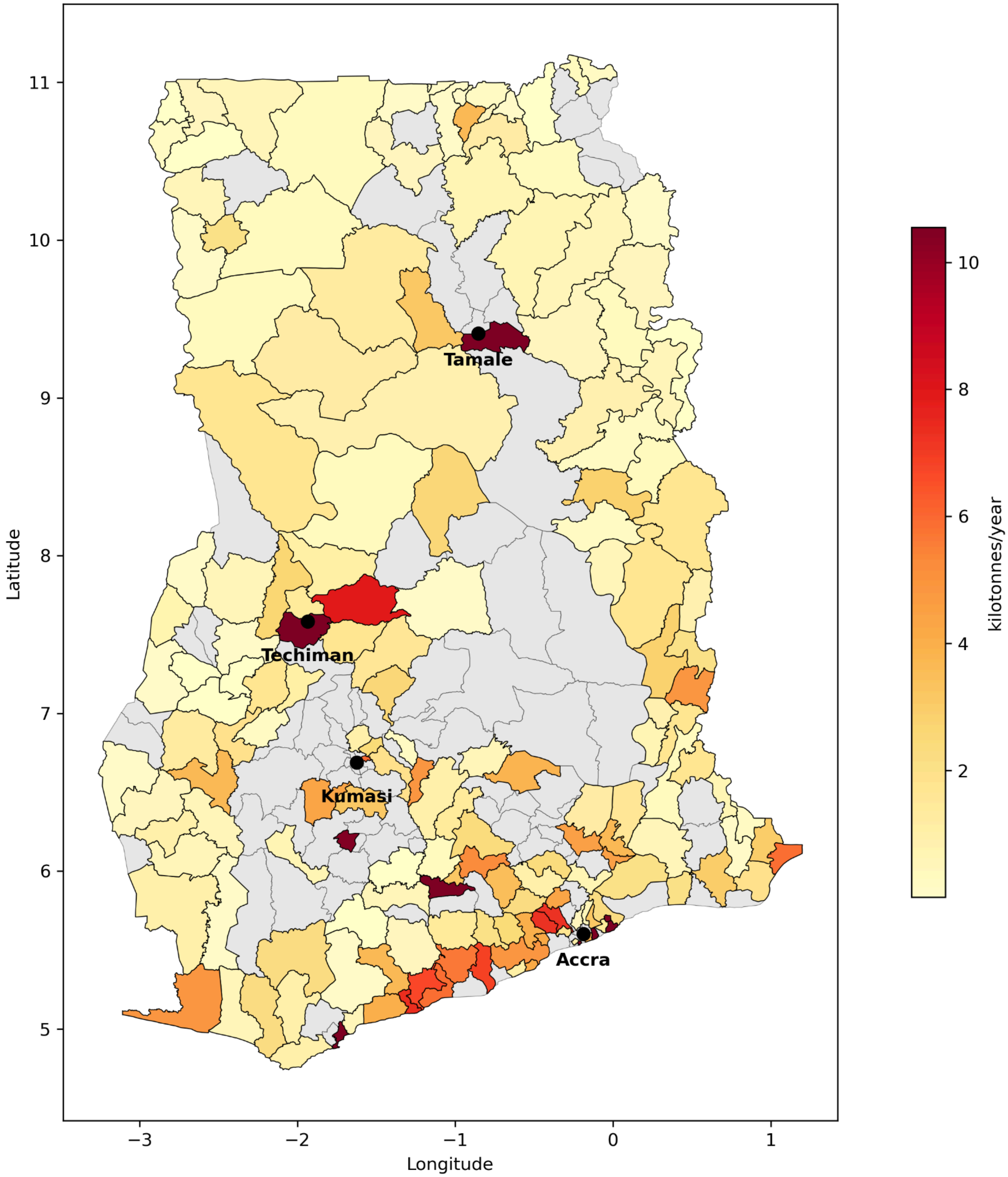

Figure 3. Annual charcoal demand from primary and secondary users for covered districts in our 2021 national survey (10).

Discussion

We developed GasPay to address liquidity constraints preventing a clean fuel transition and hopefully spur future research. We argue that DFS like GasPay can attract funding for the energy transition, creatively target vulnerable groups, and have unintended downstream effects on financial inclusion, women's economic empowerment, and well-being.

A major source of private funding for household energy is the voluntary carbon market (VCM), where efficient cookstove projects monetize emissions reductions. Pairing this carbon finance with digital financial services could further accelerate clean fuel adoption and sustained use. GasPay and other DFS for development could ensure vulnerable groups more susceptible to air pollution (e.g., pregnant women and newborns) are not only included but intentionally targeted. Finally, GasPay may advance women's financial inclusion and economic empowerment by providing first-time access to formal financial tools, greater privacy and transparency in LPG purchases, and time savings from streamlined transactions and cleaner, faster cooking.

Methods

We construct charcoal demand from our team's 2021 national household survey from questions regarding respondents' primary and secondary fuels for available districts using survey weights. We use the bootstrap approach to obtain 95% confidence intervals. We scale primary and secondary consumption values to Ghana's 2021 population. We compare carbon dioxide equivalent estimates for this consumption of charcoal and LPG using only carbon dioxide, methane, and nitrous oxide (3, 11). We draw on focus groups and interviews from our fieldwork. Details are in SI.

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